

Network Traffic Analysis for Cyber Attack Detection

A Summer Training Project Report Submitted in Partial Fulfillment for the
Award of the Degree of Bachelor of Technology in Computer Science And
Engineering (Artificial Intelligent & Machine Learning)

Submitted

To



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**DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING(ARTIFICIAL INTELLIGENCE& MACHINE
LEARNING)**

**UNITED COLLEGE OF ENGINEERING & RESEARCH,
PRAYAGRAJ, INDIA**

JULY 2026

Certificate

This is to certify that the project titled “**Network Traffic Analysis For Cyber Attack Detection**” is the genuine work carried out by **Rishabh Seth (2500101530146)** , **Akshit Pratap Singh (2500101530020)**, **Kshitij Mishra (2500100100275)** , in partial fulfillment of the requirements for the award of the **Bachelor of Technology** (Computer Science And Engineering (Artificial Intelligent & Machine Learning)) degree submitted to **Dr. A.P.J. Abdul Kalam Technical University, Lucknow**, at **United College of Engineering & Research, Prayagraj**.

The project is an authentic record of their own work carried out during the period from **June 2026** to **July 2026** under the guidance of **Dr. Snehlata, Assistant Professor**, Department of Computer Science & Engineering.

The Summer Training Project Viva-Voce Exam was held on

Signature of the Supervisor [**Dr. Snehlata**]

Signature of the Head of Department [**Dr. Vijay Kumar Dwivedi**]

Place: Prayagraj

Date:

Candidate Declaration

We declare that the project entitled “**Network Traffic Analysis For Cyber Attack Detection**”, submitted by us in partial fulfillment of the requirements for the award of the degree **Bachelor of Technology**, (Computer Science and Engineering (Artificial Intelligent & Machine Learning)) to Dr. A.P.J. Abdul Kalam Technical University, Lucknow, at United College of Engineering & Research, Prayagraj, is an authentic record of our own work carried out during the period from June 2026 to July 2026 under the guidance of **Dr. Snehlata, Assistant Professor, Department of Computer Science and Engineering**.

We further declare that the matter presented in this project has not been submitted for the award of any other degree, diploma, fellowship, or other similar title.

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Date:

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Abstract

Today, cyber attacks represent serious threats to computer systems. This project aims to identify those attacks by investigating network traffic. Our small controlled lab environment with an attacker in the form of Kali Linux operating system and a victim in the form of a Windows machine has been created in order to analyze such events.

This project had the aim to investigate how various cyber attacks appear when traveling across a network. In order to achieve this purpose, we have used a network analyzer called Wireshark which recorded all network data traffic between our two machines.

The following attacks were launched in this experiment:

- Port Scanning (to find open doors in a network)
- DDoS/SYN Flood (to flood the network and make it crash)
- Brute Force (an attack aimed at guessing passwords and break in).

The analysis of the captured network traffic through Wireshark helped us to recognize clear fingerprints of each attack. It appeared that each cyber attack has its own footprint in the network data. This project shows that careful monitoring of network traffic helps to prevent the attacks in advance.

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Chapter 1

Introduction

1.1 Cyber Security Overview

In the current world, all the activities that we perform are associated with the internet. The storage of our personal data, transfer of information, and even work activities are performed using computers, smart phones, and other networks. With the increased dependence on the internet, the risk of cyber threats has increased.

Cybersecurity is the process of protecting these computers, networks, programs, and data from any form of malicious activity which may harm their integrity and security. Cybersecurity may be viewed as the firewall that protects the computer from any form of threat. Without proper cybersecurity measures, the personal data, passwords, credit card numbers, among other private details, can be accessed and stolen. In order to effectively protect against any kind of attacks, cybersecurity measures have to adapt to new trends in hacking.

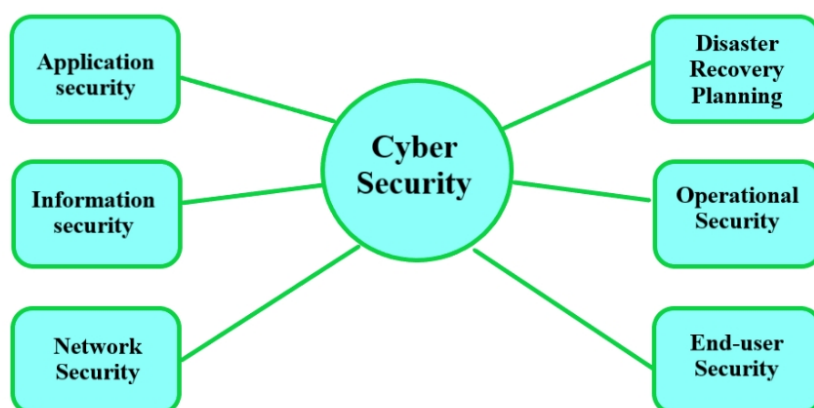


Figure 1.1: Cybersecurity Overview

1.2 Network Traffic Analysis

Before discussing how to protect a network, it is vital to know the meaning of the term "network traffic". Any time you send an email, watch videos online, or visit a web site, packets of information are sent between your computer and the computer you are communicating with on a network. These packets pass through the network millions of times within one second.

Network Traffic Analysis(NTA) is the process of monitoring, recording, and inspecting this data as it travels across the network.

1.3 Need for Cyber Attack Detection

In general, it is believed that a good password and an antivirus are sufficient for protection from any attacks. Regrettably, it is not true. In the age of cyberattacks, hackers try not to break into systems by force but search for weak spots and disguise their criminal activities with seemingly regular use of the network.

Once a hacker bypasses the basic firewall, he/she can be situated in the company's network for weeks or even months, performing all kinds of theft and espionage while nobody notices. That is why the development of a cyber attack detection tool becomes crucially important. It is not enough to prevent anything; there should also be an alerting mechanism. Analyzing network traffic at all times, we will be able to detect a cyber attack instantly, which will give us an opportunity to stop any harm.

1.4 Objectives Of the Project

The primary aim of this research is to examine methods in which we may use the data from a network to detect and counter cyber threats. The particular objectives are as follows:

- **Study of Traffic Patterns:** To learn how regular, legitimate network traffic differs from malicious attack traffic.
- **Implementation of a Detection System:** To employ the use of computer programs that monitor network traffic and alert on suspicious activities.

- **Improving Precision:** To ensure the proper detection of actual attacks without producing false positives.
- **Analysis of Security Alerts:** To assist in identifying what kind of an attack is taking place.

1.5 Scope of the Project

Cybersecurity is a large area, and it is impossible to discuss all types of threats in one project. Thus, this project will be limited by certain boundaries:

- **Environment of Attack:** For this project, we will be looking at analyzing the traffic in the simulated environment or a standard network that was already recorded beforehand instead of a live enterprise network.
- **Type of Attacks:** In this case, we will concentrate on finding out about various network attacks, such as DDoS attack (when a hacker tries to overload the system by sending it a lot of traffic) or intrusion attempt.
- **Tools used:** The project will be limited only by the use of standard network tools for analysis and monitoring that will help us analyze data packets and alert the user of possible problems. It will not include development of any automatic blocking tool, such as firewall.

Chapter 2

Methodology

This chapter describes the architecture design, process flow, and features of the Network Analysis and Real Time Threat Detection system that has been proposed. This system has been designed to consume raw traffic data, extract lightweight telemetry, create baseline behavior and cross-verify anomalies to produce alerts in real time.

2.1 System Methodology

The execution of this project will be done according to an established methodology that ensures that the analysis carried out is correct with regard to both normal and attacked networks. This begins with setting up a virtualized environment in which the connectivity between the attacker and the targeted system is checked.

1. Baseline Generation:

Traffic is collected to understand the normal functioning of the network.

2. Attack Simulation:

Three different vectors of attack, namely Port Scanning, DoS SYN Flooding, and SMB Brute Force attacks are launched through professional pen testing software.

3. Analysis & Identification:

All simulations are done along with real-time packet capturing through Wireshark. The display filters are used to distinguish packets that are malicious.

4. Documenting:

All phases are documented with relevant illustrations (packet captures and statistical graphs).

2.2 Flowchart

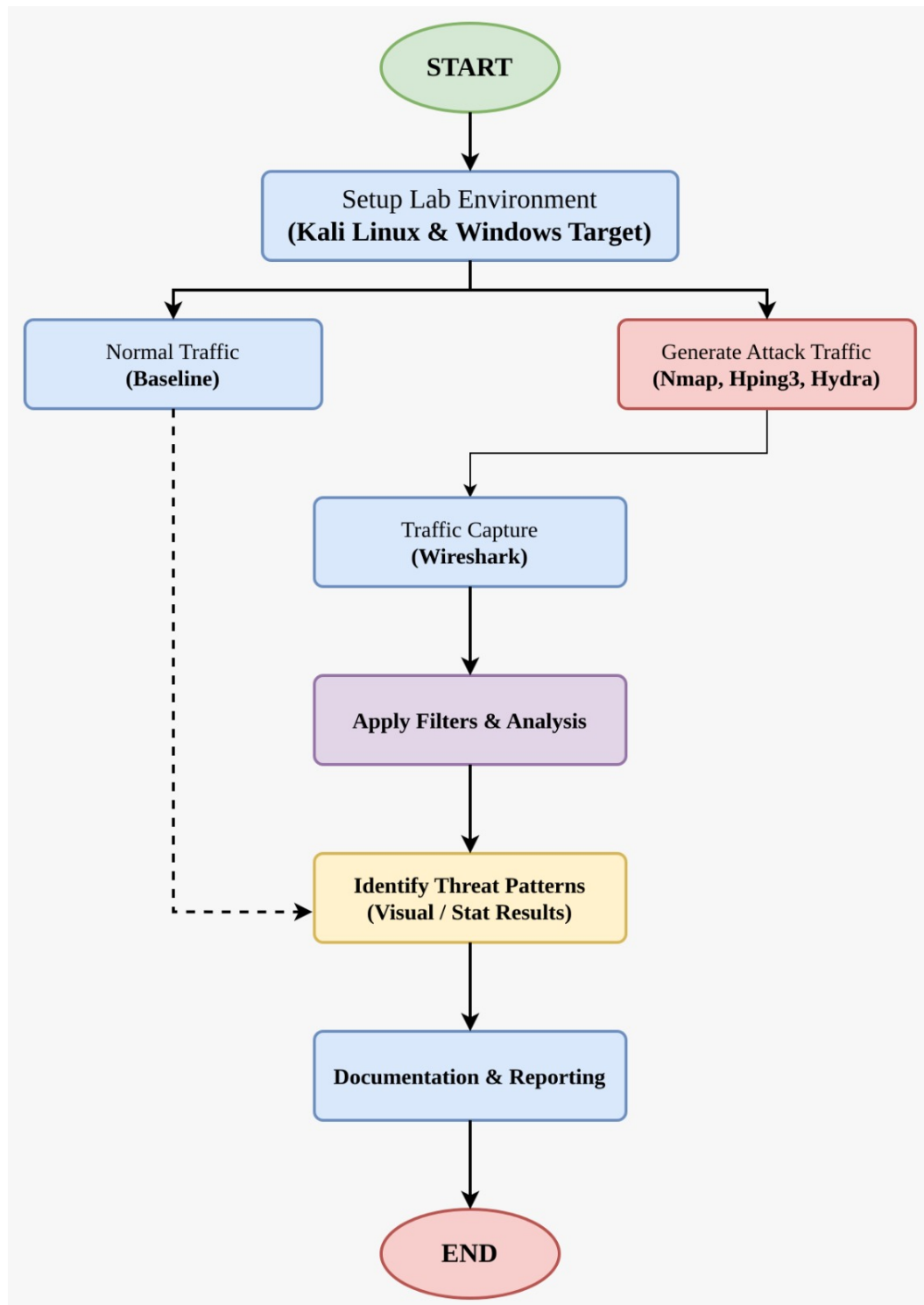


Figure 2.1: Methodology.

2.3 Tools and Technologies Used

The tools and technologies below were used to perform the network traffic analysis and simulate different cyberattacks:

For conducting this project and analyzing network traffic, specific software tools and environments were used. These tools helped us set up our lab, perform simulated attacks, and capture the data effectively. The following table provides a summary of the technologies utilized:

Category	Tools Used	Purpose
Virtualization	VirtualBox / VMware	Setting up isolated lab environment
OS	Kali Linux	Attack simulation environment
Analysis	Wireshark	Packet capture & traffic analysis
Attack Tools	Nmap, Hping3, Hydra	Simulating specific cyber attacks

Table 2.1: Tools and Technologies Used

2.3.1 Virtualization Environment:

The Kali Linux virtual machine was hosted on Oracle VirtualBox (or VMware).

2.3.2 Operating Systems:

- **Attacker Machine:** Kali Linux (based on Debian distribution). Chosen because it comes pre-installed with security/penetration testing tools.
- **Target Machine:** Windows OS, which is used as the victim machine for performing the attacks.

2.4 Network Analysis Tool:

- **Wireshark:** It was used for deep packet inspection, live traffic monitoring, and analysis of the packets at packet level in order to find signs of threats.

2.4.1 Attack Simulation Tools:

- **Nmap:** It was used to perform network reconnaissance and port scanning attacks (SYN stealth scan).

- **Hping3:** Used to perform network traffic simulation attacks (DoS-SYN flood attack).
- **Hydra:** Used to perform brute force attacks on the SMB service.

2.4.2 Utility Tools:

- **Ping (ICMP):** To check the connectivity of the host (Target) and VM (Attacker).

2.5 Detection Techniques

A detection technique refers to a process or strategy that can be employed in the identification, detection, and analysis of suspicious activities, cyber attacks, or security threats in a computer system or network.

2.5.1 Signature-based Detection

Signature-based detection recognizes cyber attacks through comparison of network traffic or files against a database containing attack signatures.

2.4.1.1 How it works?

In case traffic matches any known attack signature, an alert is generated.

2.4.1.2 Strength

It can quickly and accurately detect known attacks.

2.4.1.3 Weakness

It cannot detect new or unknown attacks.

2.5.2 Anomaly-based Detection

Anomaly-based detection involves learning the normal activity of a network and identifying any activities that do not fit in the pattern.

2.4.2.1 How it works ?

In case of abnormal or unusual activities in a network, an alert is raised.

2.4.2.2 Strength

Can detect unknown or zero-day attacks.

2.4.2.3 Weakness

Can generate false positives.

2.5.3 Rule-based Detection

Rule-based detection involves detecting threats according to defined security rules.

2.2.3.1 How it works ?

When there is a match between defined security rules and network activity, an alert is generated.

2.4.3.2 Strength

Easy to implement and customize.

2.4.3.3 Weakness

Only detects the attacks covered by defined rules.

Chapter 3

Implementation

In this project, the emphasis will be on Network Traffic Analysis for Cyber Attack Detection. The main goal would be to capture and analyze network packets in order to differentiate between baseline traffic and malicious behavior. The experimental environment will involve a simulated environment where network packets can be easily captured during cyber attacks

3.1 Lab Enviornmental Setup

“To perform this network traffic analysis, a controlled virtual environment was created. This ensures that the testing is safe and does not affect any external networks.”

- **Attacker Machine:** Kali Linux (Virtual Machine).
- **Target Machine:** Windows OS (Virtual Machine).
- **Analysis Tool:** Wireshark installed on the host/attacker system.
- **Connectivity:** Both machines were connected via the same virtual network adapter to allow packet transfer.

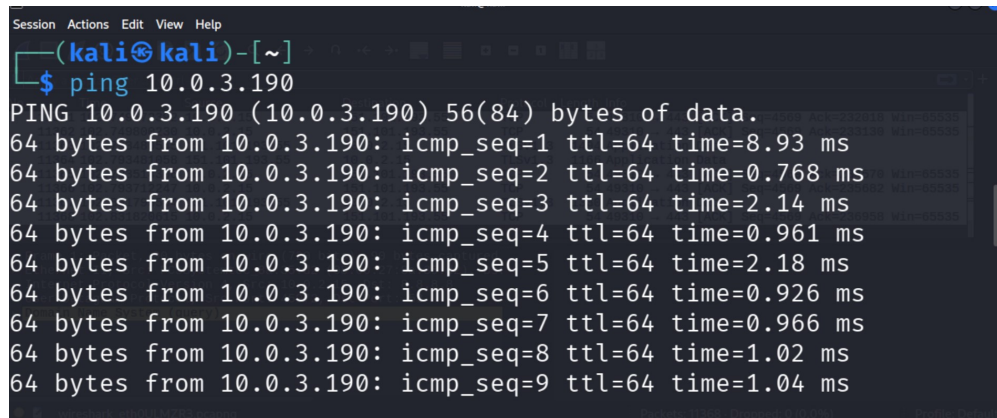
3.1.1 Baseline Traffic Analysis (Normal Traffic)

- **Objective:** To confirm the systems are connected and record “Normal” network behavior.
- **Tools used:** Ping , wireshark

1. Connectivity check (Ping Test)

Before we start any attacks, we first need to make sure that the Attacker computer (Kali Linux) and the Target computer (Windows) are connected correctly. To test this, we used a simple 'Ping' command.

- **What we did:** We sent a ping request from Kali Linux to the Windows IP address.
- **Result:** The Windows machine sent a reply back. This confirms that the two computers can talk to each other on the network.



```
Session Actions Edit View Help
(kaliⓈkali)-[~]
$ ping 10.0.3.190
PING 10.0.3.190 (10.0.3.190) 56(84) bytes of data.
64 bytes from 10.0.3.190: icmp_seq=1 ttl=64 time=8.93 ms
64 bytes from 10.0.3.190: icmp_seq=2 ttl=64 time=0.768 ms
64 bytes from 10.0.3.190: icmp_seq=3 ttl=64 time=2.14 ms
64 bytes from 10.0.3.190: icmp_seq=4 ttl=64 time=0.961 ms
64 bytes from 10.0.3.190: icmp_seq=5 ttl=64 time=2.18 ms
64 bytes from 10.0.3.190: icmp_seq=6 ttl=64 time=0.926 ms
64 bytes from 10.0.3.190: icmp_seq=7 ttl=64 time=0.966 ms
64 bytes from 10.0.3.190: icmp_seq=8 ttl=64 time=1.02 ms
64 bytes from 10.0.3.190: icmp_seq=9 ttl=64 time=1.04 ms
```

Figure 3.1: ping_test

2. Baseline Traffic Capture

- In order to define how the normal network traffic looks like, we captured the traffic while carrying out various operations on the computer. It allows us to set a benchmark so that we could quickly identify any abnormal activities.

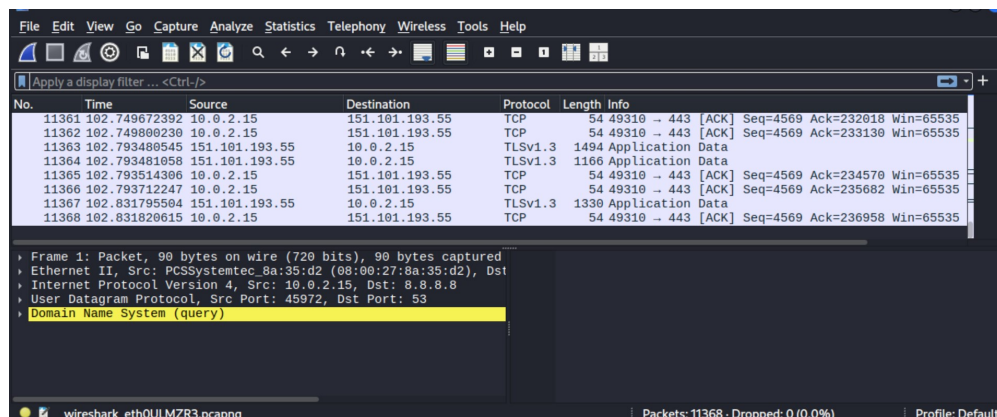


Figure 3.2: normal_traffic

Captured traffic included the following types of protocols:

- **DNS (Domain Name System):** This protocol is similar to an online directory that enables conversion of website names to IP addresses. The presence of this protocol indicates that the computer user is surfing the Internet in a usual manner.

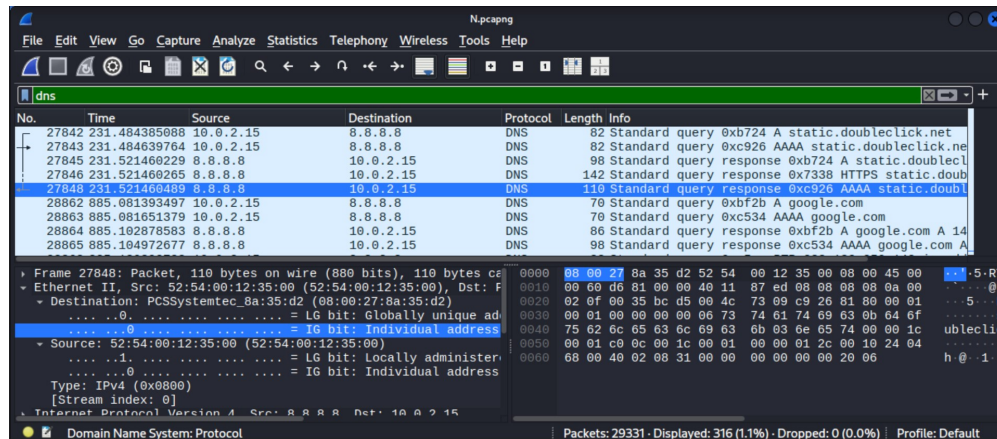


Figure 3.3: dns filter

- **TCP (Transmission Control Protocol):** Packets of this type ensure that the data exchange takes place without problems. This is a regular protocol in terms of browsing the websites.

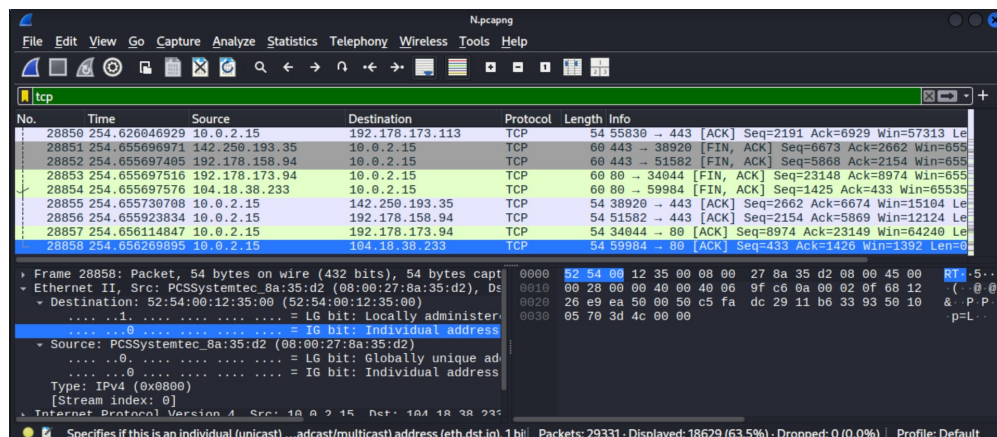


Figure 3.4: TCP Filter

- **ICMP (Internet Control Message Protocol):** Messages of this type help in verifying network integrity. This is a regular way of verifying network connections.

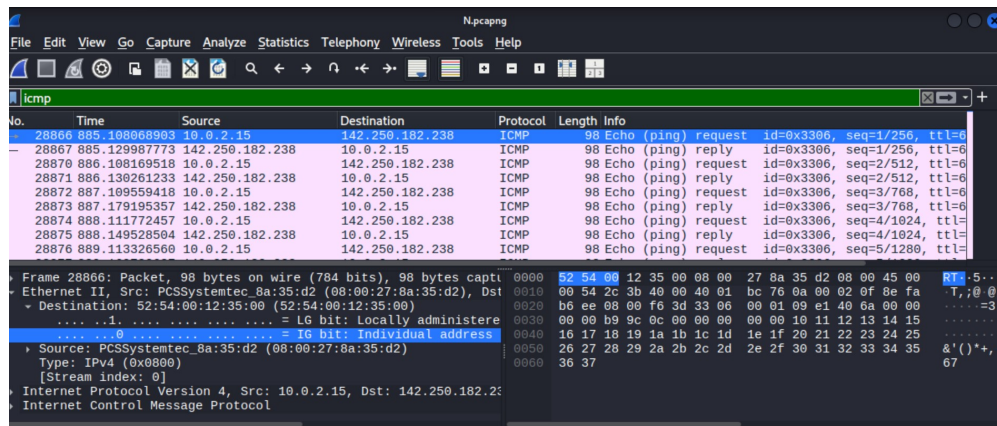


Figure 3.5: ICMP filter

- **Observation:** None of the strange or suspicious protocols was identified in this network traffic capture. There was no unusual traffic growth and connections. Thus, the benchmark traffic is considered to be regular. This is very important as we will have a benchmark to compare other traffic captures with.

3.1.2 Analysis of Port Scanning Attack

- **Objective:** To find out which ports (doors) are open on the target machine.
- **Tools Used:** Nmap.
- **Procedure:** We ran the command `nmap -sS 10.0.2.15` on the Kali terminal to perform a Stealth Scan.

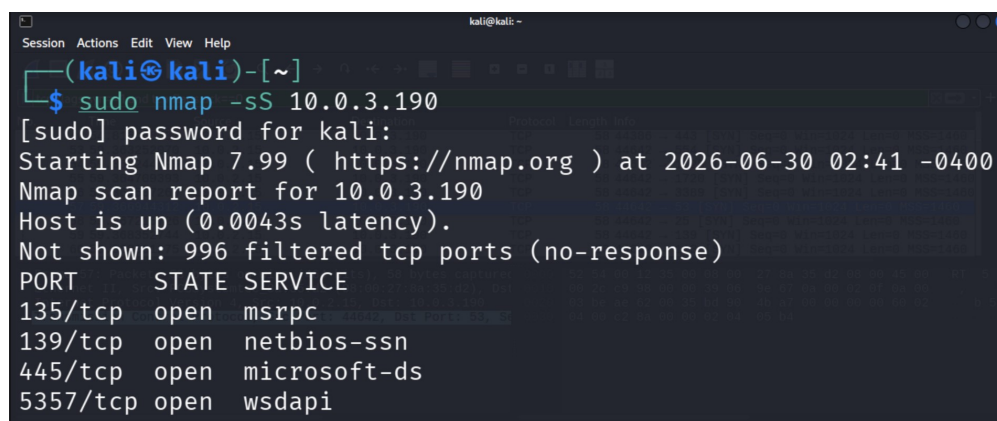


Figure 3.6: Nmap

- **Observation:** The capture shows a lot of ‘SYN’ packets going to the target. This confirms that the scanner is checking which ports are active.

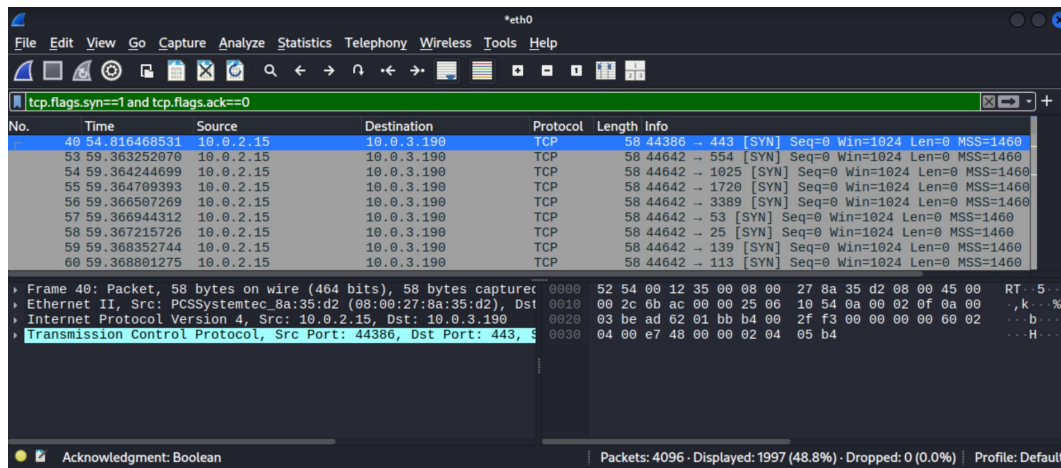


Figure 3.7: Wireshark port scan

3.1.3 Denial of Service (DoS) - SYN Flood Attack Analysis

- **Objective:** To test if the target system crashes when it receives too much traffic at once.
- **Tools Used:** Hping3.
- **Procedure:** We ran the command `sudo hping3 -S --flood -V 10.0.3.190`. We stopped it after 10 seconds to avoid crashing the virtual lab.
- **Denial of Service (DoS)** attacks seek to impair the proper function of a target server, service, or network through overloading them with excessive amounts of internet traffic. During this stage, a SYN Flood attack was created with the use of hping3 application.

```

kali@kali: ~
$ sudo hping3 -S --flood -V 10.0.3.190
[sudo] password for kali:
using eth0, addr: 10.0.2.15, MTU: 1500
HPING 10.0.3.190 (eth0 10.0.3.190): S set, 40 headers + 0 data bytes
hping in flood mode, no replies will be shown
^C
— 10.0.3.190 hping statistic —
21582 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms

```

Figure 3.8: DDoS Terminal

- SYN Flood utilizes the standard TCP three-way handshake procedure. An attacker sends multiple requests quickly to synchronize the connection in order to overload a target's resources so that the target is unable to process legitimate traffic. An attacker doesn't send the final ACK and leaves the target system with half-open connections that eat up its memory.

Wireshark was actively collecting the network data during the simulation. With the use of the display filter `tcp.flags.syn==1`, the massive amount of synchronization requests sent to the target IP from the attacker's IP became visible.

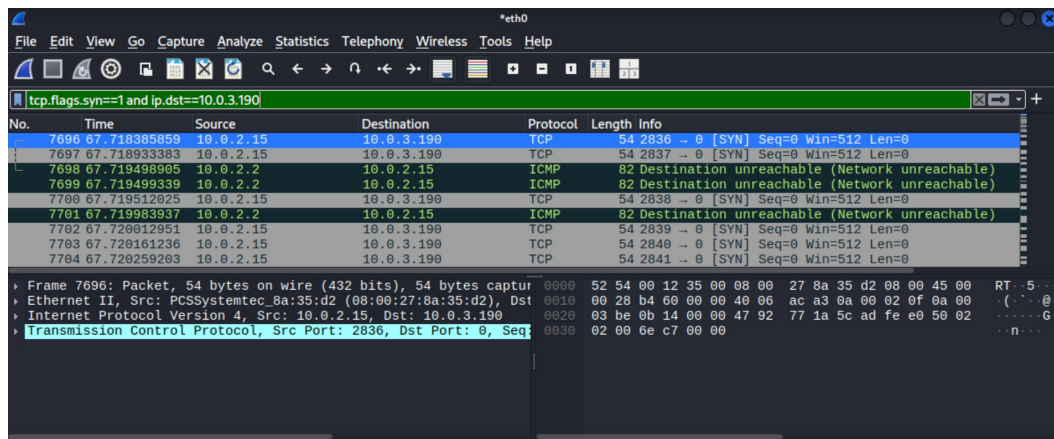


Figure 3.9: DDoS Wireshark

- In order to have a more graphical representation of how the volumetric DoS attack looks like, Wireshark's I/O graph was used.

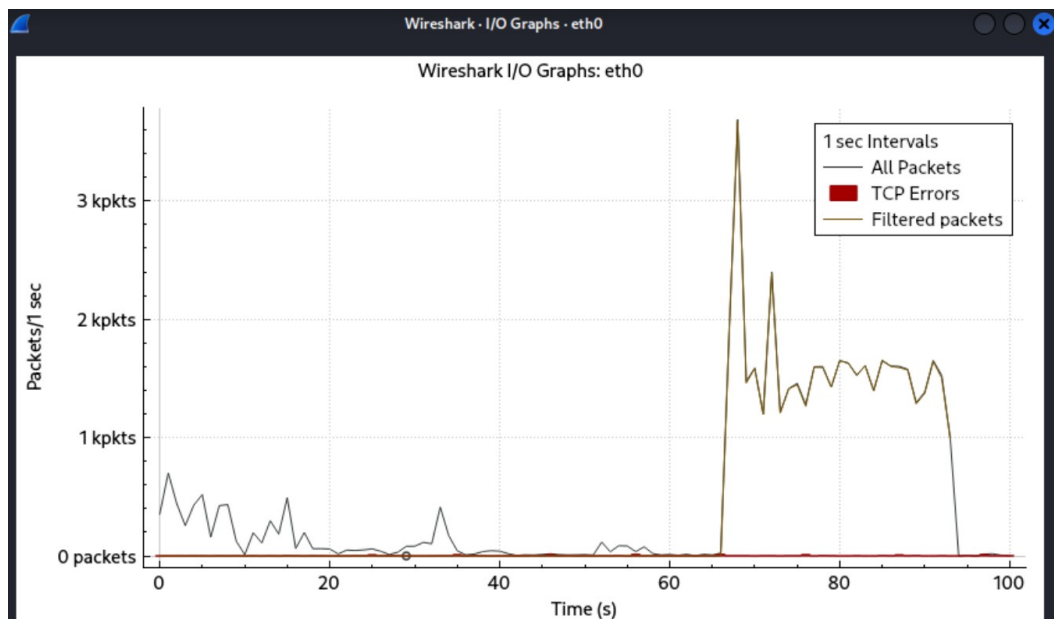


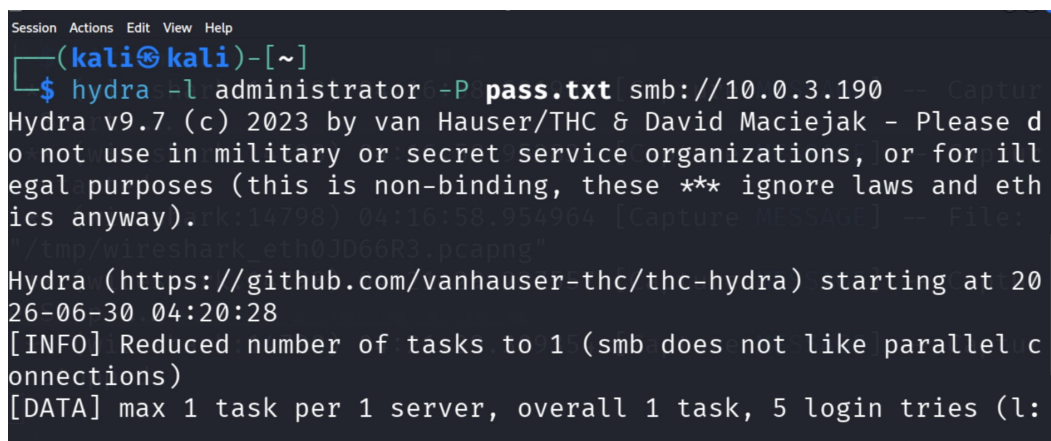
Figure 3.10: DDoS Graph

- **Observation:** Wireshark shows a massive number of SYN packets in a very short time. This flooding of requests can make the target system slow or unresponsive.

The results of the analysis of I/O Graph show a sharp increase in the rate of packet transmissions during the same time when the program hping3 is running. In addition, the result of the analysis of the captured packets proves the continuous flow of the SYN packets sent to the target machine (10.0.3.190) from one single source machine (10.0.2.15) with no ACK packets following those.

3.1.4 Brute Force Attack Analysis

- **Ojective:** To guess the login password for the target's SMB service.
- **Tool Used:** Hydra
- **Procedure:** We used the command `hydra -l administrator -P pass.txt smb://10.0.3.190` to try multiple password one by one.
- Brute Force attacks are cryptographic techniques where attackers attempt to guess login credentials, encryption keys, or hidden web pages by trial and error. They comprise automatic software attempting numerous combinations of user IDs and passwords one after another. In the present case, a brute force attack utilizing dictionary was performed on the SMB (Server Message Block) service operating on port 445 of the target machine from the attacker machine using the Hydra network login hacking utility.



```

Session Actions Edit View Help
(kali㉿kali)-[~]
$ hydra -l administrator -P pass.txt smb://10.0.3.190
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-06-30 04:20:28
[INFO] Reduced number of tasks to 1 (smb does not like parallel connections)
[DATA] max 1 task per 1 server, overall 1 task, 5 login tries (l:

```

Figure 3.11: bruteforce_{terminal}

- Wireshark was employed to capture the specific signatures of the network created during this process of authentication flooding. The packet capture was then filtered by using the display filter `tcp.port == 445`.

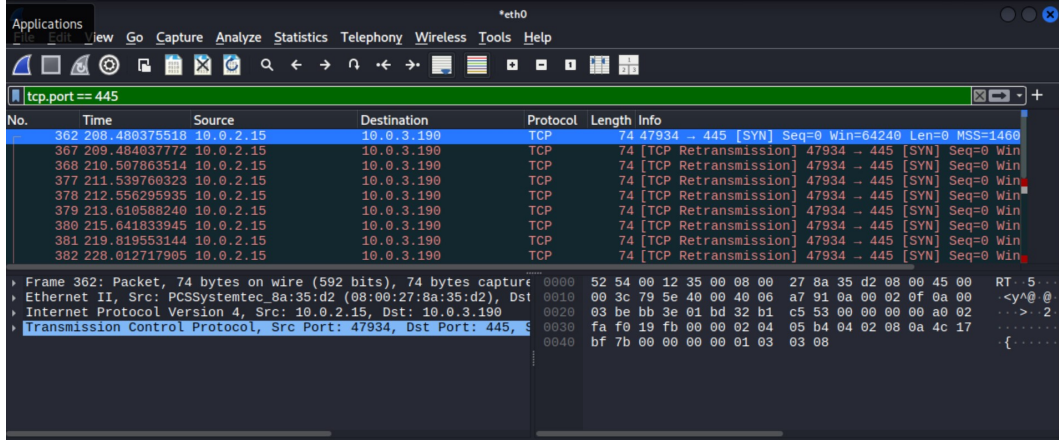


Figure 3.12: *bruteforce_{wireshark}*

- **Observations:** The network traffic analysis shows a unique fingerprint that characterizes a brute force attack. The presence of an unusually high number of connections over the TCP protocol made to port 445 over a period of just a few seconds by the attacking IP (10.0.2.15) is noticed. In a real-world scenario, where logging is turned on, this represents multiple login attempts and `STATUS_LOGON_FAILURE` errors. The high number of connections made to the authentication interface in a short span of time is a strong evidence of a brute force attack.

Chapter 4

Results and Analysis

4.1 Overview

In this project, we successfully set up a virtual network lab and simulated different types of cyber attacks. By capturing traffic with Wireshark, we were able to see exactly how these attacks look on a network.

4.2 Summary of Findings

We tested four main scenarios. Below is a summary of what we observed in each case:

Attack Phase	Tool Used	Observation	Result
Phase 1	None (Baseline)	Standard DNS/TCP packets	Normal Activity
Phase 2	Nmap	Rapid sequence of SYN packets	Port Scan
Phase 3	Hping3	Massive flood of SYN requests	DoS Attack
Phase 4	Hydra	Repeated login failures	Brute Force

Table 4.1: Summary of Findings

4.3 Analysis

The analysis shows that every cyber attack leaves a "digital fingerprint."

- **Scanning:** It looks like someone is knocking on every door of a house to see which one opens.
- **DoS Attack:** It is like sending thousands of letters to one mailbox at once, making it impossible to read any real mail.
- **Brute Force:** It is like someone trying different keys on a lock over and over again until it opens.

By filtering the traffic in Wireshark, we proved that we can separate "Normal" traffic from "Attack" traffic. This proves that simple network monitoring tools can be very effective for security.

Chapter 5

Conclusion and Future Scope

5.1 Conclusion

This project showed that understanding network traffic is very important for cybersecurity. By using Kali Linux to perform attacks and Wireshark to monitor them, we learned how to spot threats by looking at network data. We successfully proved that:

1. **Cyber attacks have specific patterns (signatures).**
2. **Wireshark is a powerful tool to capture and analyze these patterns.**
3. **Establishing a baseline of “normal” behavior is the first step to catching hackers.**

Overall, this project provided a hands-on experience in detecting threats, which is a fundamental skill for building strong network security defenses.

5.2 Future Scope

While this project was done manually, there is a lot of room for improvement in the future:

- **Automation:** Instead of manually filtering traffic, we can write Python scripts to automatically detect these attacks and send an alert.
- **Machine Learning:** We could train a computer model to recognize these attack patterns automatically without human help.

- **Real-time Defense:** This system could be connected to a Firewall, so that as soon as an attack is detected, the Firewall automatically blocks the attacker's IP address.

Chapter 6

References

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